

a bit by assuming nonlinearity. Take what is generally called the “butterfly in India” paradigm from the discussion of Lorenz’s discovery in the previous chapter. As we have seen, a small input in a complex system can lead to nonrandom large results, depending on very special conditions. A single butterfly flapping its wings in New Delhi may be the certain *cause* of a hurricane in North Carolina, though the hurricane may take place a couple of years later. However, *given the observation of a hurricane in North Carolina*, it is dubious that you could figure out the causes with any precision: there are billions of billions of such small things as wing-flapping butterflies in Timbuktu or sneezing wild dogs in Australia that could have caused it. The process from the butterfly to the hurricane is greatly simpler than the reverse process *from the hurricane to the potential butterfly*.

Confusion between the two is disastrously widespread in common culture. This “butterfly in India” metaphor has fooled at least one filmmaker. For instance, *Happenstance* (a.k.a. *The Beating of a Butterfly’s Wings*), a French-language film by one Laurent Firode, meant to encourage people to focus on small things that can change the course of their lives. Hey, since a small event (a petal falling on the ground and getting your attention) can lead to your choosing one person over another as a mate for life, you should focus on these very small details. Neither the filmmaker nor the critics realized that they were dealing with the backward process; there are trillions of such small things in the course of a simple day, and examining all of them lies outside of our reach.

Once Again, Incomplete Information

Take a personal computer. You can use a spreadsheet program to generate a random sequence, a succession of points we can call a history. How? The computer program responds to a very complicated equation of a nonlinear nature that produces numbers that seem random. The equation is very simple: if you know it, you can predict the sequence. It is almost impossible, however, for a human being to reverse engineer the equation and predict further sequences. I am talking about a simple one-line computer program (called the “tent map”) generating a handful of data points, not about the billions of simultaneous events that constitute the real history of the world. In other words, even if history were a nonrandom series generated by some “equation of the world,” as long as reverse engineering such an equation does not seem within human possibility, it should be deemed random and not bear the name “deterministic chaos.” Historians should stay away from chaos theory and the difficulties of reverse engineering except to discuss general properties of the world and learn the limits of what they can’t know.